

UNIT 3 Part B

Visual Analytics and Data Visualization - Prescriptive Analytics: Building Linear Optimization models, - Implementing Linear Optimization models on spreadsheets, -Solving Linear Optimization models- CASE STUDIES.

Visual Analytics

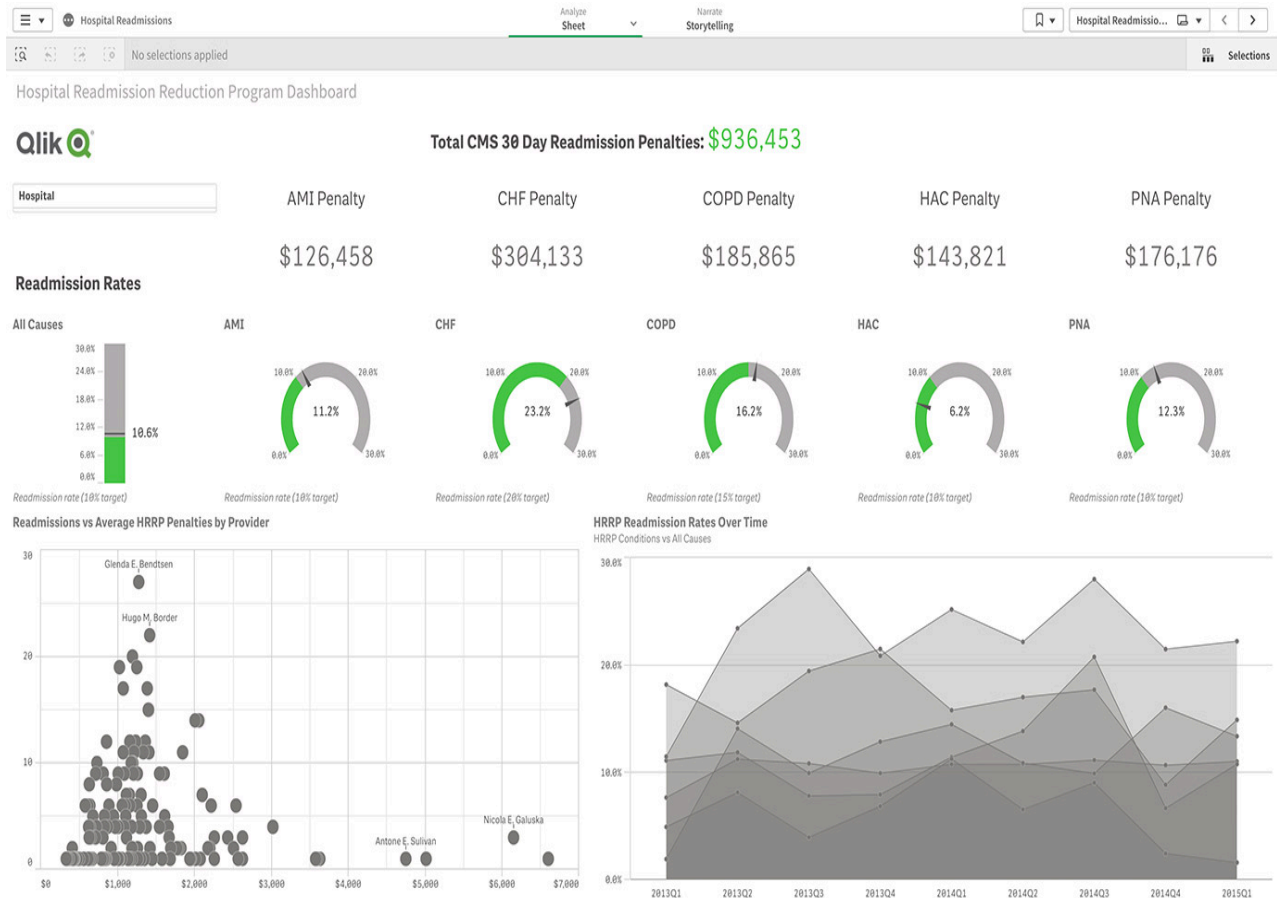
What is Visual Analytics?

Visual analytics is the use of sophisticated tools and processes to analyze datasets using visual representations of the data. Visualizing the data in graphs, charts, and maps helps users identify patterns and thereby develop actionable insights. These insights help organizations make better, data-driven decisions.

Visual Analytics Benefits

Sometimes confused with data visualization, **visual analytics isn't simply a matter of representing data graphically**. Modern, interactive visual analytics makes it easy to combine data from multiple sources and deeply analyze the data directly within the visualization itself. Plus, AI and machine learning algorithms can offer recommendations to help guide your exploration.

Ultimately, **visual analytics helps you turn massive data sets into business insights** which can have a major positive impact on your organization.

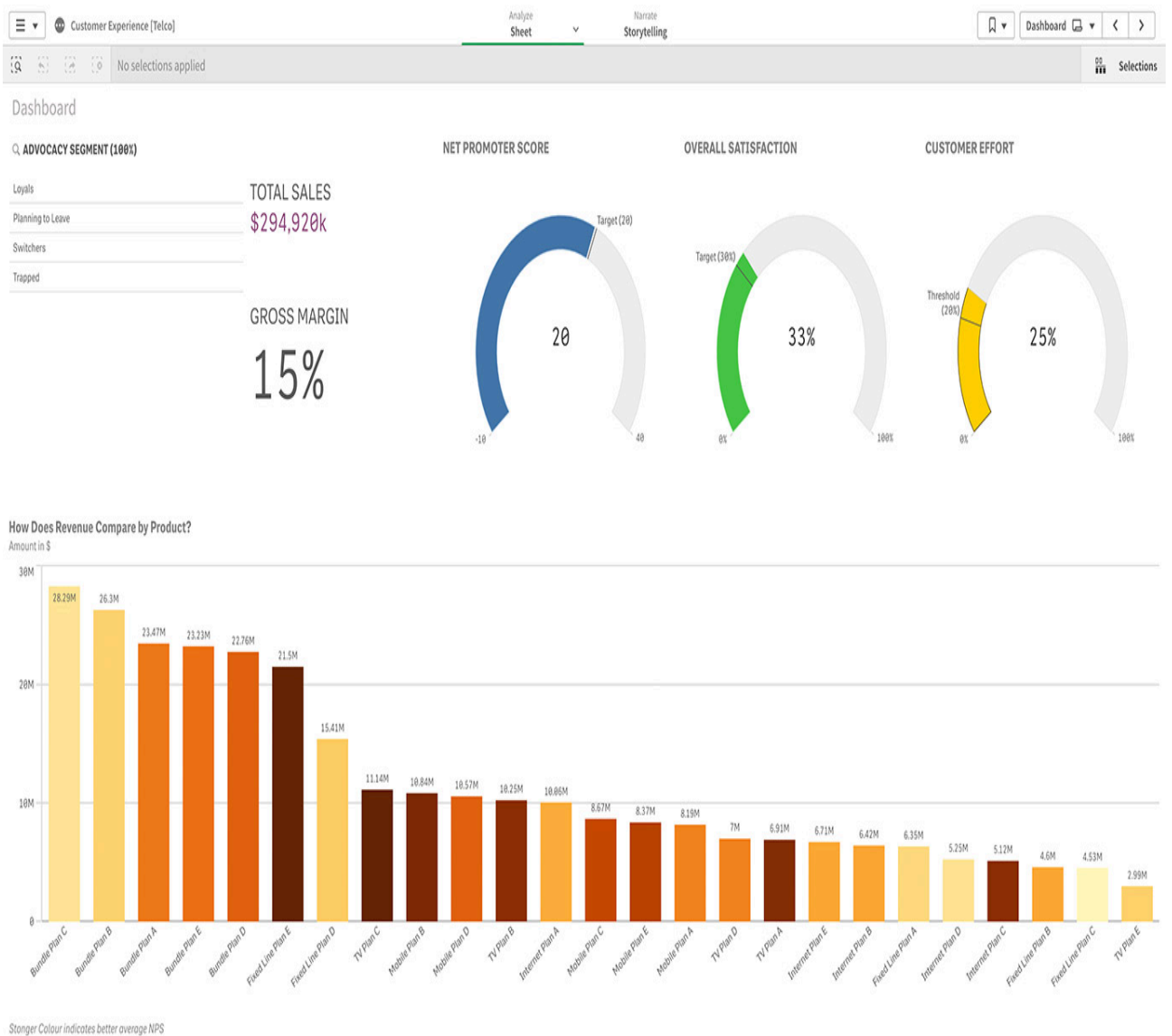


1. Share findings and track progress: Interactive reports and dashboards help users track, organize and share key performance indicators across an organization.
2. Make faster decisions: Users can understand data insights much more quickly by seeing and working with data sets when they are in a visual format.
3. Explore data more easily: Self-service analytics tools which allow users to interact with data in a visual context allows them to discover hidden relationships and patterns in the data without relying on help from IT.
4. Promote data literacy: Making data easier to work with and understand democratizes data analytics, getting more people across an organization involved.

Visual Analytics Examples based on KPIs relevant to specific roles.

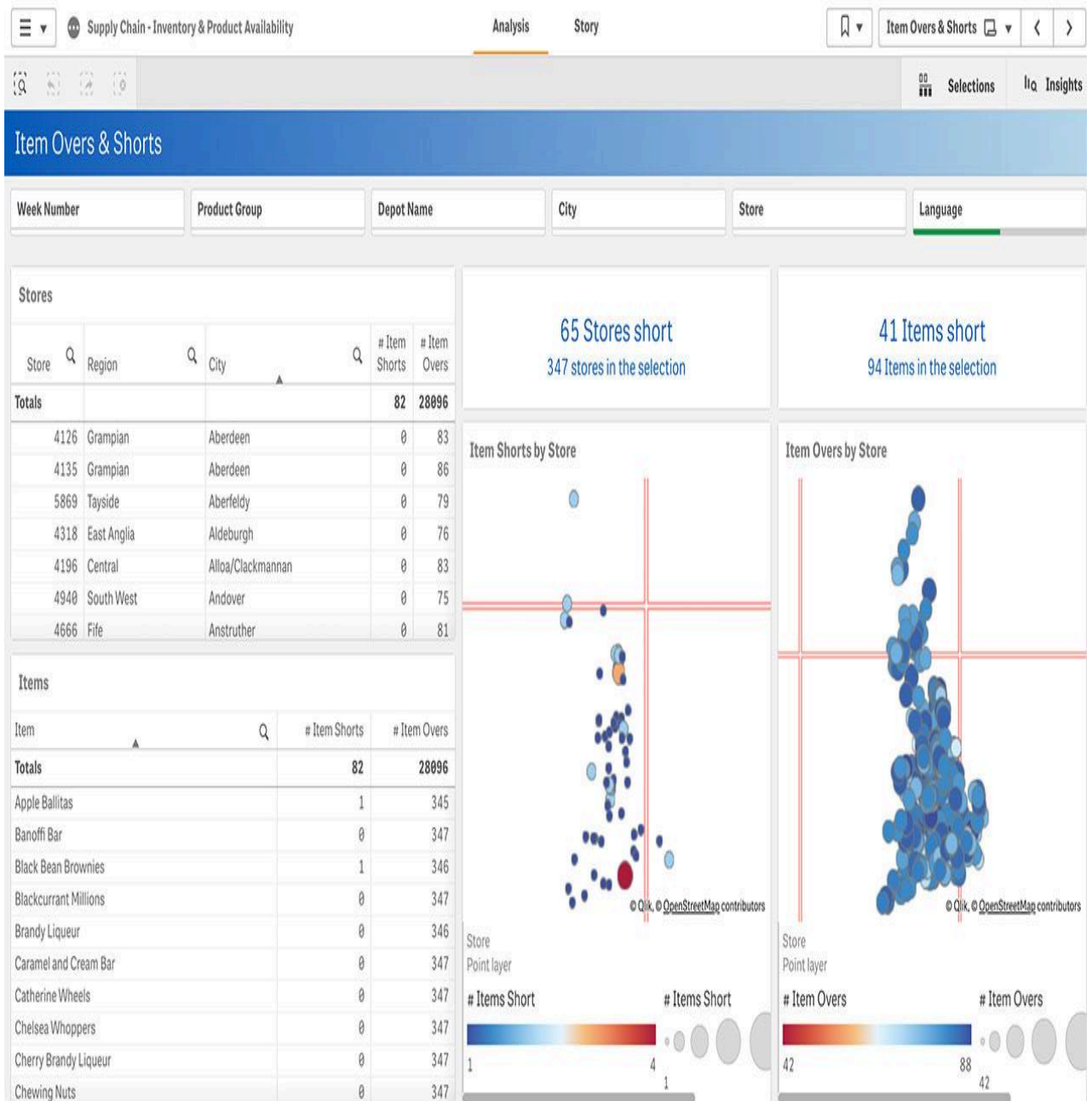
Marketing

In this example, visual analytics helps a marketer boost ROI by letting them see and understand each phase of the customer life cycle. This is made possible by connecting data from the CRM, advertising tools, and web analytics platform.



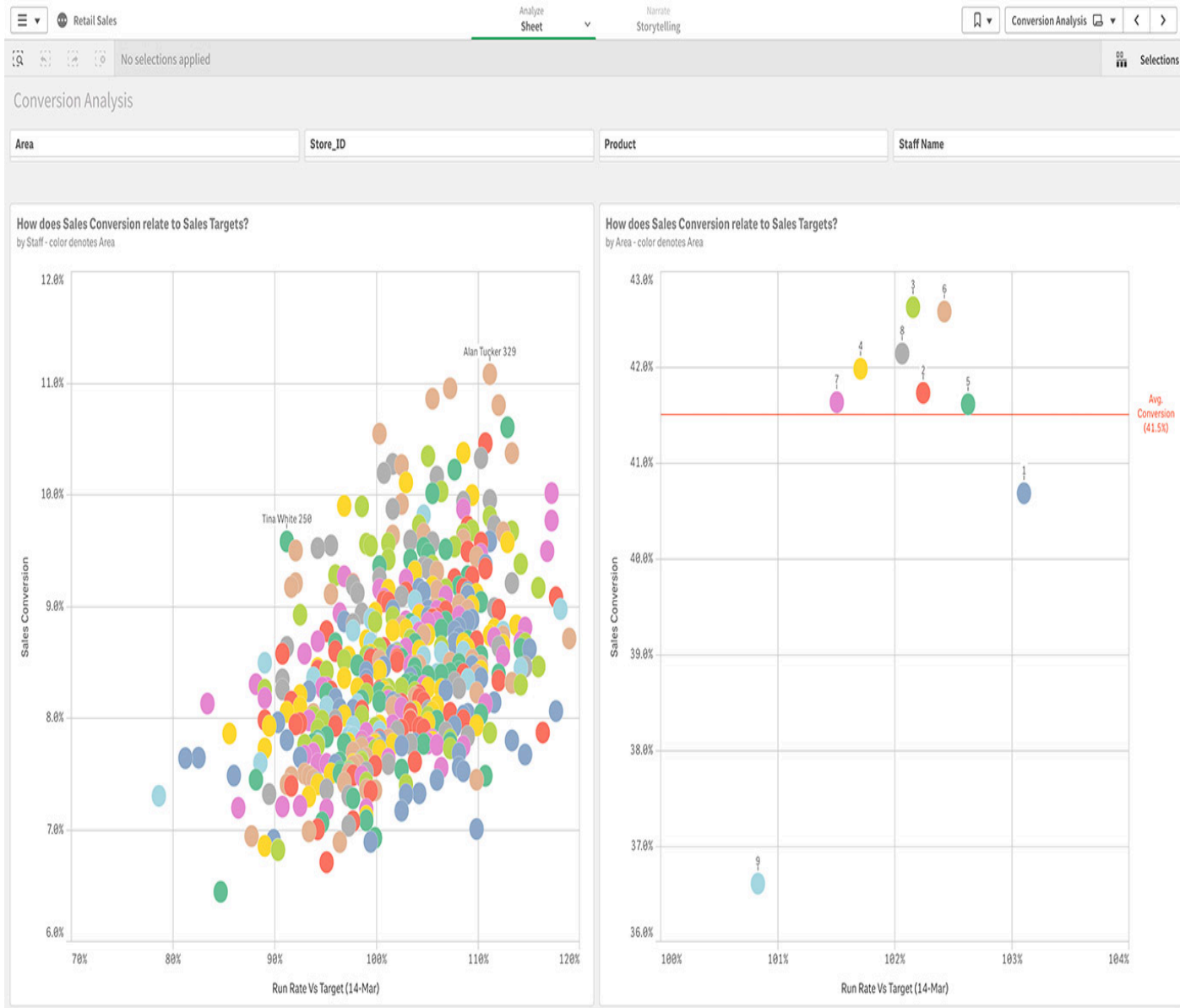
Supply Chain

Big data visual analytics can help supply chain managers immediately find connections between complex, disparate data sources by surfacing KPIs and allowing interactive exploration.



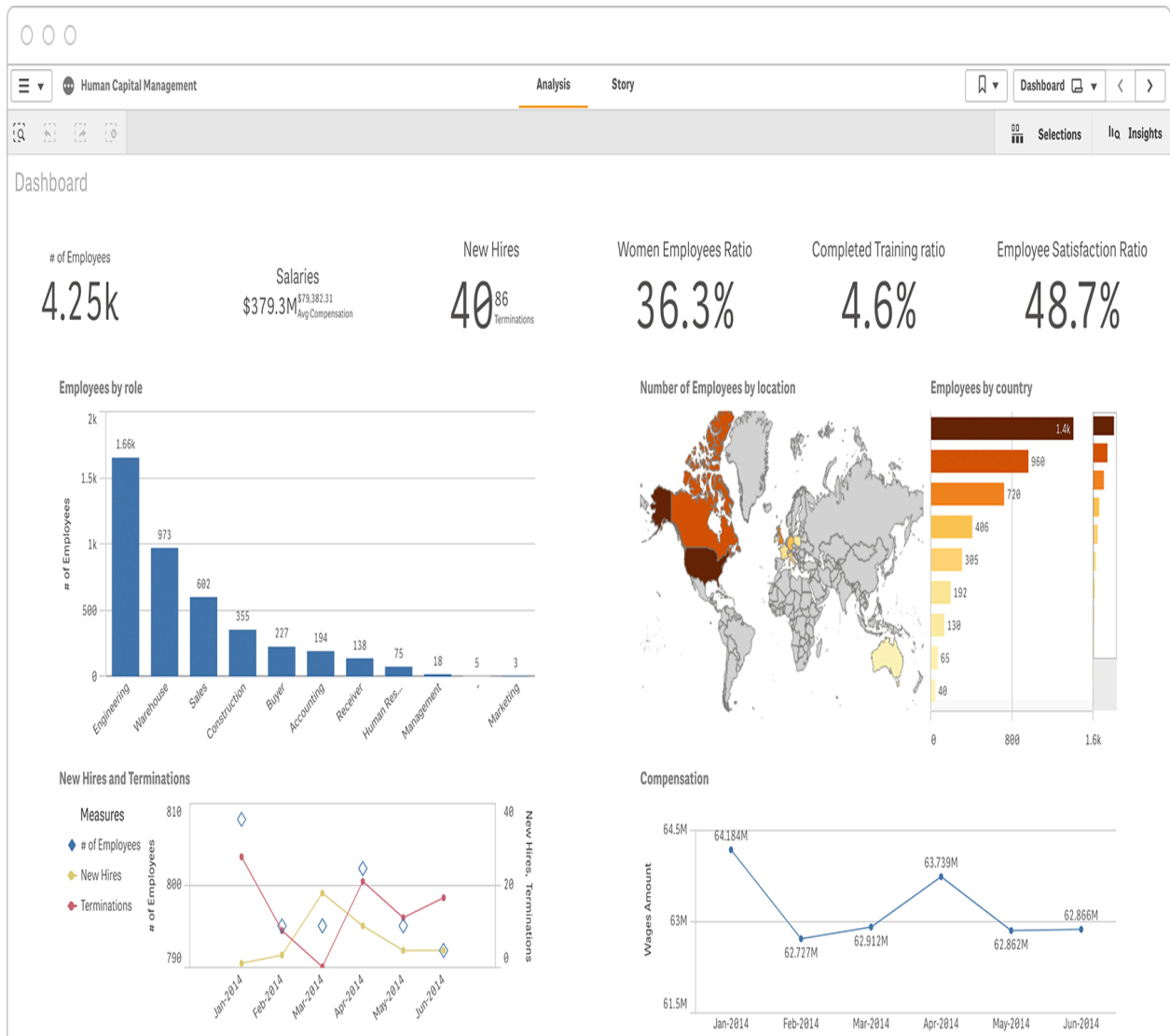
Sales

Sales managers benefit from the clean, organized presentation of sales data to increase revenue, improve forecasting, and identify key trends.



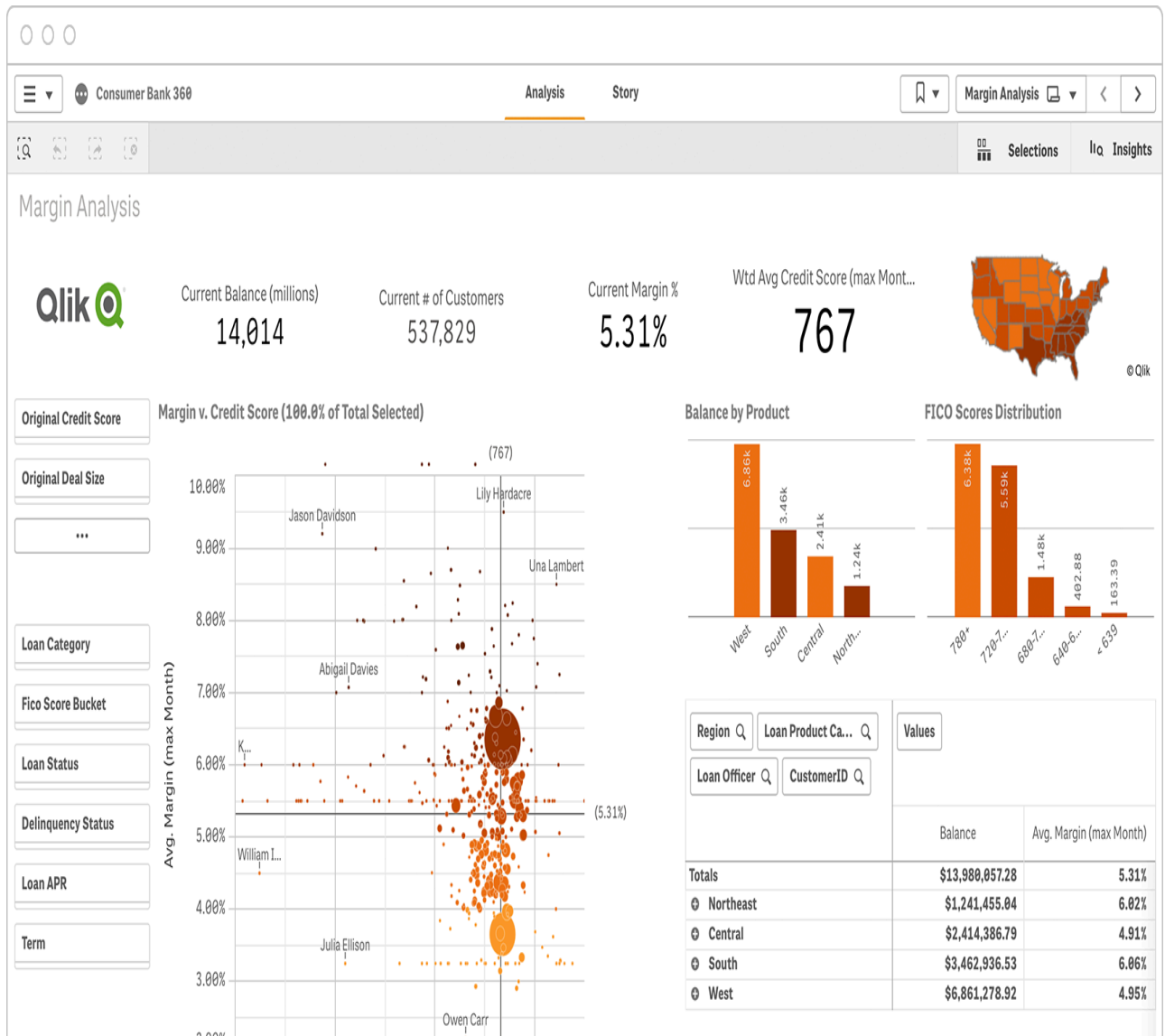
Human Resources

Visualizations can provide HR executives the ability to review and analyze KPIs in a single dashboard. This example shows data on employees by role and by location, plus key data such as gender ratio and training progress.



Finance

In this example, a loan manager at a consumer bank can explore how different regions, product and loan officers perform over time and drive the biggest impact on revenue and margins.



IT Visual Analytics

IT managers can leverage data analytics and visualization to better anticipate upcoming technology needs and identify underused systems and applications.

Dashboard

High Priority Cases

72

Medium Priority Cases

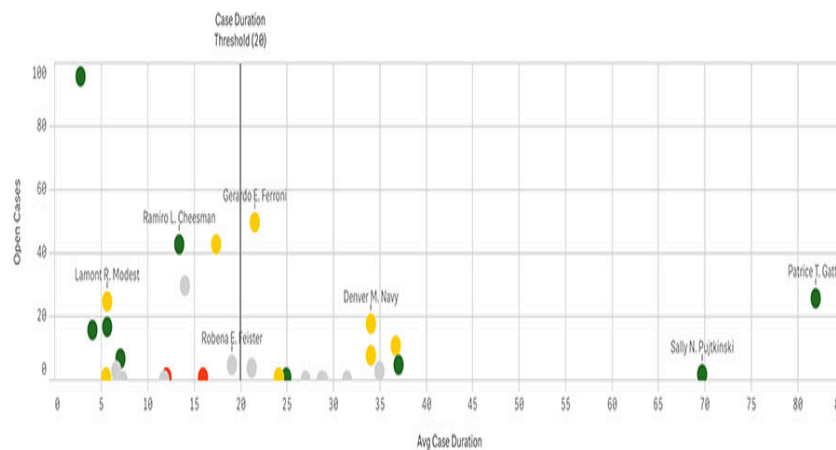
142

Low Priority Cases

203

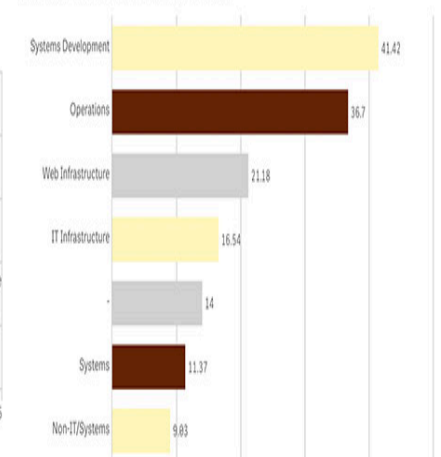
Open Cases vs Duration

* color represents the priority with the most open cases

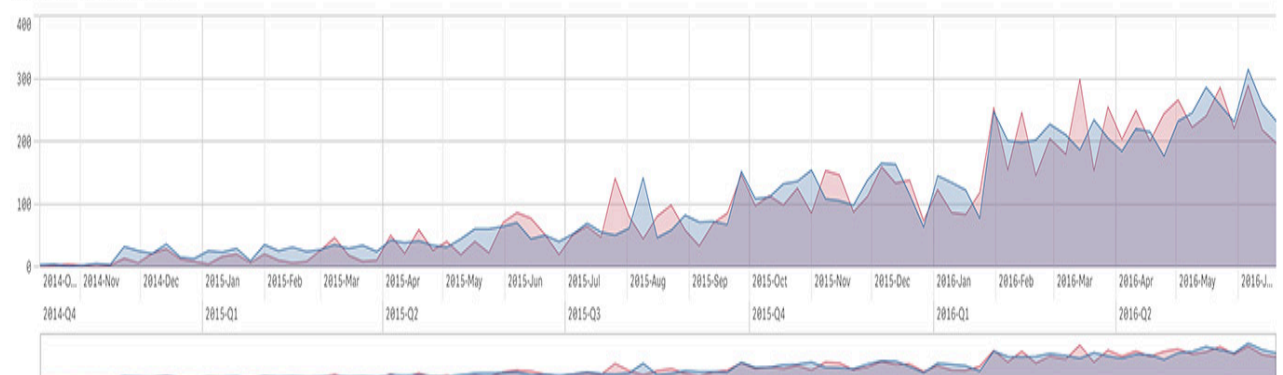


Avg Case Resolution Time (Days)

* color represents the priority with the most open cases



Open & Resolved Cases Over Time



Visual Analytics Best Practices: The following key best practices

1. **Define goals:** Before beginning - define specific goals for usual data analysis work. What specific questions are to be answered?
2. **Integrate and manage the data**
3. **Your source data needs to be transformed into clean, business-ready information.** You'll need to combine and replicate data from a variety of sources and then bring it into

standardized formats stored in a repository such as a data lake or data warehouse.

4. Simplify visualizations
5. Choose the right visual technique to present your story in the simplest way possible.

Visual Analytics Software Capabilities: The best data analytics tools include the following capabilities:

1. Integrate and manage big data: The best tools can combine and manage data from multiple sources to give users the full picture.
2. Allow free exploration: Your tool should allow you to freely explore data in whatever direction your intuition leads. Static charts and linear drill-downs make it harder to answer your questions.
3. Suggest visuals with AI: Augmented analytics recommends visualizations which help even novice users build their own analytics views and discover hidden insights.
4. Embed everywhere: Your software should let users explore visual data analysis from within whatever websites, portals, apps and business processes they use to make decisions.

Data Visualization

Data Visualization: Data visualization is the graphical representation of information and data in a pictorial or graphical format(Example: charts, graphs, and maps). Data visualization tools provide an accessible way to see and understand trends, patterns in data and outliers. Data visualization tools and technologies are essential to analyze massive amounts of information and make data-driven decisions. The concept of using pictures to understand data has been used since centuries. General types of data visualizations are Charts, Tables, Graphs, Maps, Dashboards.

Based on	Data Visualization
Definition	Data visualization is the graphical representation of information and data in a pictorial or graphical format.
Benefits	Identify areas that need attention or improvement Clarity which factors influence customer behavior Helps understand which products to places where Predict sales volumes
Used for	The goal of the data visualization is to communicate information clearly and efficiently to users by presenting them visually.

Relation	Data visualization helps to get better perception.
Industries	Data Visualization technologies and techniques are widely used in Finance, Banking, Healthcare, Retailing etc
Tools	Plotly, DataHero, Tableau, Dygraphs, QlikView, ZingCHhart etc.
Platforms	Big data processing, Service management dashboards, Analysis and design.
Techniques	Data visualization can be static or interactive.
Performed by	Data Engineers

Difference between data visualization and visual analytics

The term data visualization typically refers to the graphical depiction of data, or representing data in bubble charts, heat maps, and other visuals to help people better understand the patterns, relationships, trends, and other meaningful insights in datasets. Visual analytics refers to the use of an analytics program to perform advanced analysis of complex datasets and allow users to explore and interact with dynamic visualizations.

What Is Prescriptive Analytics?

Prescriptive analytics is a branch of data analytics that uses predictive models to suggest actions to take for optimal outcomes. Forecasting the load on the electric grid over the next 24 hours is an example of *predictive analytics*, whereas deciding how to operate power plants based on this forecast represents *prescriptive analytics*.

Prescriptive analytics relies on optimization and rules-based techniques for decision making.

Optimization techniques such as linear programming, integer programming, and nonlinear programming play an important role in prescriptive analytics, since they enable a set of decisions to be made in an optimal way. These techniques are applied to a model, which represents the decisions to be made, constraints on the decisions, and an objective for comparing the decisions.

Rules-based techniques including inference engines, scorecards, and decision trees are used in prescriptive analytics to make a decision such as choosing to shut down equipment for maintenance when sensor readings exceed thresholds, or accepting a financial transaction when

its score is high enough. Prescriptive analytics starts with data and ends with decisions.

Building Linear Optimization models:

Linear programming is a method of optimising operations with some constraints. The main objective of linear programming is to maximize or minimize the numerical value. It consists of linear functions which are subjected to the constraints in the form of linear equations or in the form of inequalities. Linear programming is considered an important technique that is used to find the optimum resource utilisation. The term “linear programming” consists of two words as linear and programming. The word “linear” defines the relationship between multiple variables with degree one. The word “programming” defines the process of selecting the best solution from various alternatives.

Steps involved in building a linear optimization model

1. Understand the problem
2. Describe the objective
3. Define the decision variables
4. Write the objective function
5. Describe the constraints
6. Write the constraints in terms of the decision variables
7. Add the nonnegativity constraints
8. Maximize

What is Linear Programming?

Linear programming (LP) or Linear Optimisation may be defined as the problem of maximizing or minimizing a linear function that is subjected to linear constraints. The constraints may be equalities or inequalities. The optimisation problems involve the calculation of profit and loss. Linear programming problems are an important class of optimisation problems, that helps to find the feasible region and optimise the solution in order to have the highest or lowest value of the function.

In other words, linear programming is considered as an optimization method to maximize or minimize the objective function of the given mathematical model with the set of some requirements which are represented in the linear relationship. The main aim of the linear programming problem is to find the optimal solution.

Linear programming is the method of considering different inequalities relevant to a situation and calculating the best value that is required to be obtained in those conditions. Some of the assumptions taken while working with linear programming are:

- The number of constraints should be expressed in the quantitative terms
- The relationship between the constraints and the objective function should be linear
- The linear function (i.e., objective function) is to be optimised

Components of Linear Programming

The basic components of the LP are as follows:

- Decision Variables
- Constraints
- Data
- Objective Functions

Characteristics of Linear Programming

The following are the five characteristics of the linear programming problem:

Constraints – The limitations should be expressed in the mathematical form, regarding the resource.

Objective Function – In a problem, the objective function should be specified in a quantitative way.

Linearity – The relationship between two or more variables in the function must be linear. It means that the degree of the variable is one.

Finiteness – There should be finite and infinite input and output numbers. In case, if the function has infinite factors, the optimal solution is not feasible.

Non-negativity – The variable value should be positive or zero. It should not be a negative value.

Decision Variables – The decision variable will decide the output. It gives the ultimate solution of the problem. For any problem, the first step is to identify the decision variables.

Linear Programming Problems

The Linear Programming Problems (LPP) is a problem that is concerned with finding the optimal value of the given linear function. The optimal value can be either maximum value or minimum value. Here, the given linear function is considered an objective function. The objective function can contain several variables, which are subjected to the conditions and it has to satisfy the set of linear inequalities called linear constraints. The linear programming problems can be used to get the optimal solution for the following scenarios, such as manufacturing problems, diet problems, transportation problems, allocation problems and so on.

Methods to Solve Linear Programming Problems

The linear programming problem can be solved using different methods, such as the

1. graphical method,
2. simplex method, or
3. by using tools such as R,
4. open solver etc.

Linear Programming Applications

A real-time example would be considering the limitations of labours and materials and finding the best production levels for maximum profit in particular circumstances. It is part of a vital

area of mathematics known as optimisation techniques. The applications of LP in some other fields are

- Engineering – It solves design and manufacturing problems as it is helpful for doing shape optimisation
- Efficient Manufacturing – To maximise profit, companies use linear expressions
- Energy Industry – It provides methods to optimise the electric power system.
- Transportation Optimisation – For cost and time efficiency.

Importance of Linear Programming

Linear programming is broadly applied in the field of optimisation for many reasons. Many functional problems in operations analysis can be represented as linear programming problems. Some special problems of linear programming are such as network flow queries and multi-commodity flow queries are deemed to be important to have produced much research on functional algorithm

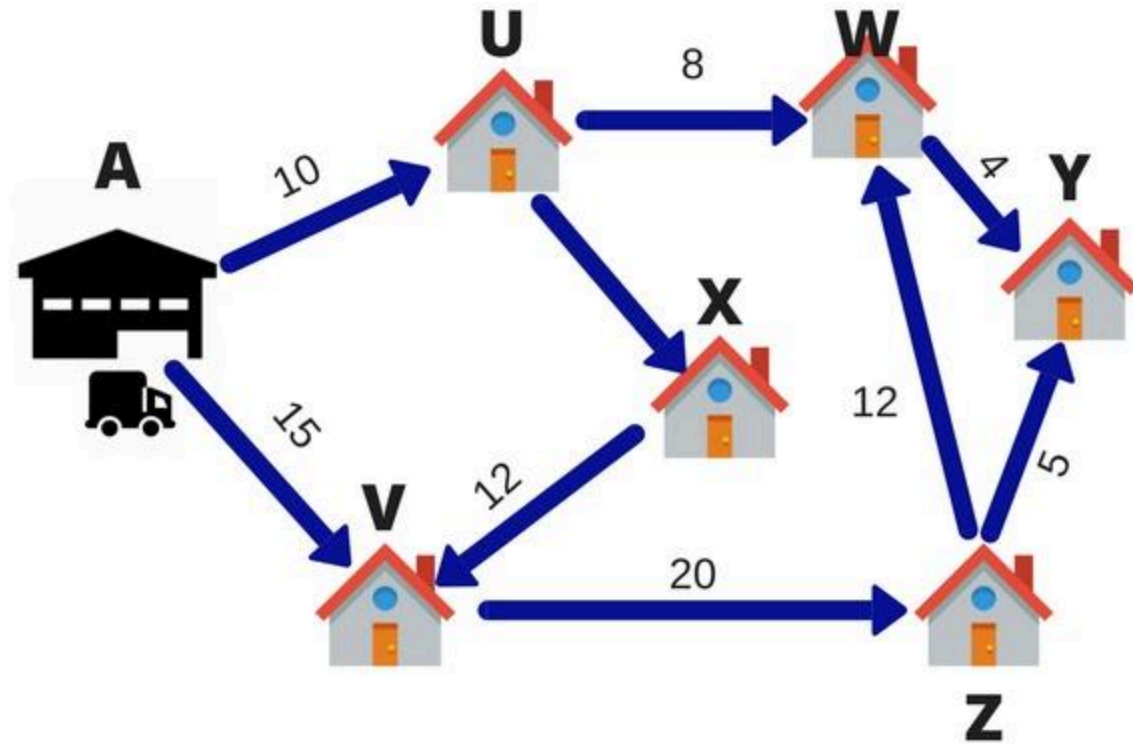
Linear Programming Video Lesson

It's important to note that linear programming is a special case of mathematical programming. It involves finding the optimal solution to a problem that involves linear relationships between the decision variables and the constraints. This method has applications in various industries and fields, from production planning to logistics optimization, and is a powerful tool in making data-driven decisions that can lead to improved efficiency and cost savings.

Solving Linear Optimization models- CASE STUDIES:

Example of a Linear Programming Problem (LPP)

Let's say a FedEx delivery man has 6 packages to deliver in a day. The warehouse is located at point A. The 6 delivery destinations are given by U, V, W, X, Y, and Z. The numbers on the lines indicate the distance between the cities. To save on fuel and time the delivery person wants to take the shortest route.



So, the delivery person will calculate different routes for going to all 6 destinations and then come up with the shortest route. This technique of choosing the shortest route is called linear programming.

In this case, the objective of the delivery person is to deliver the parcel on time at all 6 destinations. The process of choosing the best route is called Operation Research. Operation research is an approach to decision-making, which involves a set of methods to operate a system. In the above example, my system was the Delivery model.

Linear programming is used for obtaining the most optimal solution for a problem with given constraints. In linear programming, we formulate our real-life problem into a mathematical model. It involves an objective function, and linear inequalities with subject to constraints.

Is the linear representation of the 6 points above representative of the real world? Yes and No. It is an oversimplification as the real route would not be a straight line. It would likely have multiple turns, U-turns, signals and traffic jams. But with a simple assumption, we have reduced the complexity of the problem drastically and are creating a solution that should work in most scenarios.

Formulating a Problem

Let's manufacture some chocolates...

Example: Consider a chocolate manufacturing company that produces only two types of chocolate – A and B. Both the chocolates require Milk and Choco only. To manufacture each unit of A and B, the following quantities are required:

- Each unit of A requires 1 unit of Milk and 3 units of Choco
- Each unit of B requires 1 unit of Milk and 2 units of Choco

The company kitchen has a total of 5 units of Milk and 12 units of Choco. On each sale, the company makes a profit of

- Rs 6 per unit A sold
- Rs 5 per unit B sold.

Now, the company wishes to maximize its profit. How many units of A and B should it produce respectively?

Solution: The first thing I'm gonna do is represent the problem in a tabular form for better understanding.

Milk	Choco	Profit per unit	
A	1	3	Rs 6
B	1	2	Rs 5
Total	5	12	

Let the total number of units produced by A be = X

Let the total number of units produced by B be = Y

Now, the total profit is represented by Z

The total profit the company makes is given by the total number of units of A and B produced multiplied by its per-unit profit of Rs 6 and Rs 5 respectively.

Profit: $\text{Max } Z = 6X + 5Y$

which means we have to maximize Z.

The company will try to produce as many units of A and B to maximize the profit. But the resources Milk and Choco are available in a limited amount.

As per the above table, each unit of A and B requires 1 unit of Milk. The total amount of Milk available is 5 units. To represent this mathematically,

$$X + Y \leq 5$$

Also, each unit of A and B requires 3 units & 2 units of Choco respectively. The total amount of

Choco available is 12 units. To represent this mathematically,
 $3X+2Y \leq 12$

Also, the values for units of A can only be integers.

So we have two more constraints, $X \geq 0$ & $Y \geq 0$

For the company to make maximum profit, the above inequalities have to be satisfied.

This is called formulating a real-world problem into a mathematical model.

Common Terminologies Used in Linear Programming

- **Decision Variables:** The decision variables are the variables that will decide my output. They represent my ultimate solution. To solve any problem, we first need to identify the decision variables. For the above example, the total number of units for A and B denoted by X & Y respectively are my decision variables.
- **Objective Function:** It is defined as the objective of making decisions. In the above example, the company wishes to increase the total profit represented by Z. So, profit is my objective function.
- **Constraints:** The constraints are the restrictions or limitations on the decision variables. They usually limit the value of the decision variables. In the above example, the limit on the availability of resources Milk and Choco are my constraints.
- **Non-negativity Restriction:** For all linear programs, the decision variables should always take non-negative values. This means the values for decision variables should be greater than or equal to 0.

The Process of Formulating a Linear Programming Problem

Let us look at the steps of defining a Linear Programming problem generically:

- Identify the decision variables
- Write the objective function
- Mention the constraints
- Explicitly state the non-negativity restriction

For a problem to be a linear programming problem, the decision variables, objective function and constraints all have to be linear functions.

If all the three conditions are satisfied, it is called a Linear Programming Problem.

Case study : Solve Linear Programs by Graphical Method

Example: A farmer has recently acquired a 110 hectares piece of land. He has decided to grow Wheat and barley on that land. Due to the quality of the sun and the region's excellent climate, the entire production of Wheat and Barley can be sold. He wants to know how to plant each variety in the 110 hectares, given the costs, net profits and labor requirements according to the data shown below:

Variety	Cost (Price/Hec)	Net Profit (Price/Hec)	Man-days/Hec
Wheat	100	50	10
Barley	200	120	30

The farmer has a budget of US\$10,000 and availability of 1,200 man-days during the planning horizon. Find the optimal solution and the optimal value.

Solution: To solve this problem, first formulate linear program.

Formulation of a Linear Problem

Step 1: Identify the decision variables

The total area for growing Wheat = X (in hectares)

The total area for growing Barley = Y (in hectares)

X and Y are my decision variables.

Step 2: Write the objective function

Since the production from the entire land can be sold in the market. The farmer would want to maximize the profit for his total produce. We are given net profit for both Wheat and Barley. The farmer earns a net profit of US\$50 for each hectare of Wheat and US\$120 for each Barley.

Objective function (given by Z) is, $\text{Max } Z = 50X + 120Y$

Step 3: Writing the constraints

1. It is given that the farmer has a total budget of US\$10,000. The cost of producing Wheat and Barley per hectare is also given.

The total cost spent by the farmer is: $100X + 200Y \leq 10,000$

2. The next constraint is the upper cap on the availability of the total number of man-days for the planning horizon. The total number of man-days available is 1200.

As per the table, the man-days per hectare for Wheat and Barley. $10X + 30Y \leq 1200$

3. The third constraint is the total area present for plantation.

The total available area is 110 hectares: $X + Y \leq 110$

Step 4: The non-negativity restriction

The values of X and Y will be greater than or equal to 0. $X \geq 0, Y \geq 0$

Solving an LP Through the Graphical Method

Since we know that $X, Y \geq 0$. We will consider only the first quadrant.
To plot for the graph for the above equations,
Simplify all the equations.

$100X + 200Y \leq 10,000$ can be simplified to $X + 2Y \leq 100$ by dividing by 100.

$10X + 30Y \leq 1200$ can be simplified to $X + 3Y \leq 120$ by dividing by 10.

The third equation is in its simplified form, $X + Y \leq 110$.

Plot the first 2 lines on a graph in the first quadrant (like shown below)

The optimal feasible solution is achieved at the point of intersection where the budget & man-days constraints are active.

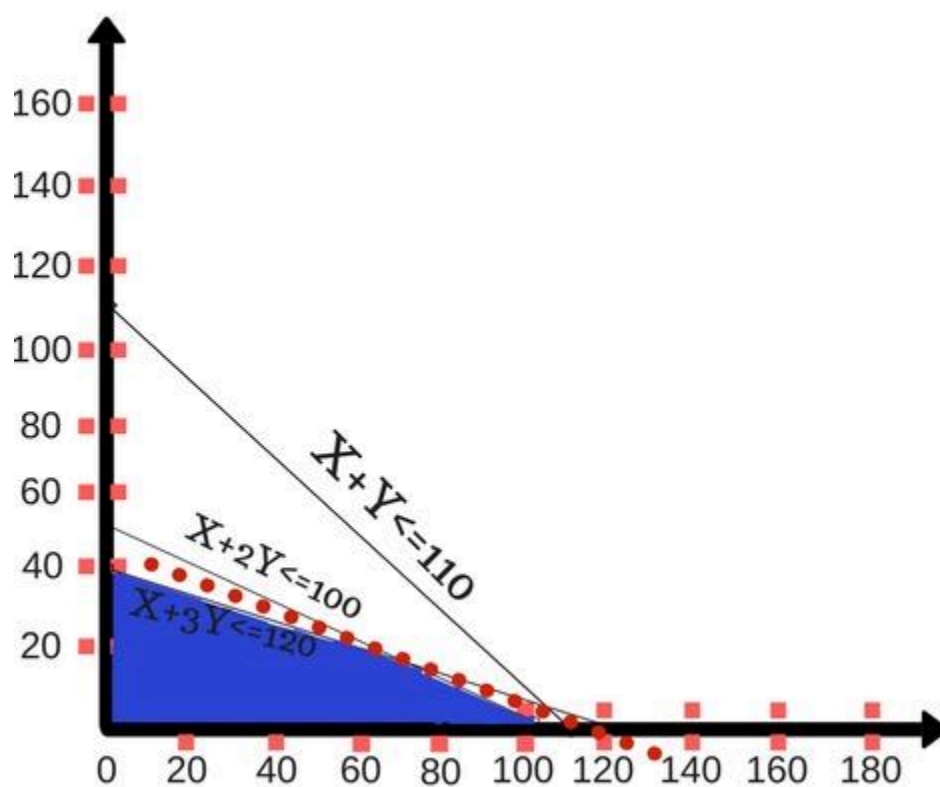
This means the point at which the equations $X + 2Y \leq 100$ and $X + 3Y \leq 120$ intersect gives us the optimal solution.

The values for X and Y which gives the optimal solution is at $(60, 20)$.

To maximize profit the farmer should produce Wheat and Barley in 60 hectares and 20 hectares of land respectively.

The maximum profit the company will gain is,

$$\text{Max } Z = 50 * (60) + 120 * (20) = \text{US\$5400}$$



Case study: **Solve Linear Program Using R**

R is an open-source tool that is very popular among the data scientists for essential data science tasks. Performing linear programming is very easy and we can attain an optimum solution in very few steps.

Come let's learn.

Example: A toy manufacturing organization manufactures two types of toys A and B. Both the toys are sold at Rs.25 and Rs.20 respectively. There are 2000 resource units available every day from which the toy A requires 20 units while toy B requires 12 units. Both of these toys require a production time of 5 minutes. Total working hours are 9 hours a day. What should be the manufacturing quantity for each of the pipes to maximize the profits?

Here:

The objective function is: $\text{Max. } Z = 25x + 20y$

where x are the units of pipe A, y are the units of pipe B

Constraints:

$$20x + 12y \leq 2000$$

$$5x + 5y \leq 540$$

Code:

```
install.packages("lpSolve")
library(lpSolve)
#Setting the coefficients of decision variables
objective.in=c(25,20)
#Constraint Matrix
const.mat=matrix(c(20,12,5,5),nrow = 2,byrow = T)
#defining constraints
const_time=540 #in minutes
const_res=2000
#RHS for constraints
const.rhs=c(const_res,const_time)
#Direction for constraints
const.dir=c("<=", "<=")
#Finding the optimum solution
opt=lp(direction =
'max',objective.in,const.mat,const.dir,const.rhs)
summary(opt)
#Objective values of x and y
opt$solution
#Value of objective function at optimal point
opt$objval
```

Output

```
summary(opt)
```

```

Length Class Mode
direction 1 -none- numeric
x.count 1 -none- numeric
objective 2 -none- numeric
const.count 1 -none- numeric
constraints 8 -none- numeric
int.count 1 -none- numeric
int.vec 1 -none- numeric
bin.count 1 -none- numeric
binary.vec 1 -none- numeric
num.bin.solns 1 -none- numeric
objval 1 -none- numeric
solution 2 -none- numeric
presolve 1 -none- numeric
compute.sens 1 -none- numeric
sens.coef.from 1 -none- numeric
sens.coef.to 1 -none- numeric
duals 1 -none- numeric
duals.from 1 -none- numeric
duals.to 1 -none- numeric
scale 1 -none- numeric
use.dense 1 -none- numeric
dense.col 1 -none- numeric
dense.val 1 -none- numeric
dense.const.nrow 1 -none- numeric
dense.ctr 1 -none- numeric
use.rw 1 -none- numeric
tmp 1 -none- character
status 1 -none- numeric
> opt$solution
[1] 88 20
> opt$objval
[1] 2600

```

Therefore from the output, we see that the organization should produce 88 units of toy A and 20 units of toy B and the maximum profit for the organization will be Rs.2600.

Case Study: Solve Linear Program Using OpenSolver

In reality, a linear program can contain 30 to 1000 variables and solving it either Graphically or Algebraically is next to impossible. Companies generally use OpenSolver to tackle these real-world problems. OpenSolver is an open-source linear and optimizer for Microsoft Excel. It is an advanced version of a built-in Excel Solver.

Example: Below there is a diet chart that gives me calories, protein, carbohydrate and fat content for 4 food items. Sara wants a diet with minimum cost. The diet chart is as follows:

Food Item 1	Food Item 2	Food Item 3	Food Item 4	
Calories	400	200	150	500
Protien (in grams)	3	2	0	0
Carbohydrates (in grams)	2	2	4	4
Fat (in grams)	2	4	1	5
Cost	\$0.50	\$0.20	\$0.30	\$0.80

The chart gives the nutrient content as well as the per-unit cost of each food item. The diet has to be planned in such a way that it should contain at least 500 calories, 6 grams of protein, 10 grams of carbohydrates and 8 grams of fat.

Solution: First, formulate my linear program in a spreadsheet.

- Step 1: Identify the decision variables. Here my decision variables are the food items. Add the headers. For trial purposes, we are entering arbitrary values. Let's say, Sara consumes 3 units of Food Item 1, 0 unit of Food Item 2, 1 unit of Food Item 3 and 0 unit of Food Item 4. These are called variable cells.

	A	B	C	D	E	F
1	Decision Variables					
2		Food Item 1	Food Item 2	Food Item 3	Food Item 4	
3	Eaten	3	0	1	0	
4						

- Step 2: Now we will write our objective function. For the diet to be optimal we must have minimum cost along with required calories, protein, carbohydrate, and fat.

4						
5	Objective Function					
6		Food 1	Food 2	Food 3	Food 4	
7	Eaten	3	0	1	0	
8	Cost per unit	0.5	0.2	0.3	0.8	

In cell B7:E7 we take the reference to the number of units. And in cell B8:E8 we put the per-unit cost of each food item.

In cell B10, we want the total cost of the diet. The total cost is given by the sumproduct of the number of units eaten and per-unit cost. Sumproduct is given by $=B7*B8+C7*C8+D7*D8+E7*E8$. Let's see this in a spreadsheet.

	A	B	C	D	E	F
1	Decision Variables					
2		Food Item 1	Food Item 2	Food Item 3	Food Item 4	
3	Eaten	3	0	1	0	
4						
5	Objective Function					
6		Food 1	Food 2	Food 3	Food 4	
7	Eaten	3	0	1	0	
8	Cost per unit	0.5	0.2	0.3	0.8	
9						
10	Total Cost	1.8	=Sumproduct (B7:E7, B8:E8)			
11						

- Step 3: Now, we will enter the constraints. Column F contains a total of calories, protein, carbohydrate, and fat. The total number of calorie intake is given by sumproduct the number of food items eaten and the calorie consumed per food item. For cell F13=Sumproduct(\$B\$7:\$F\$7, B13:F13). Similarly for others. Column G gives the inequality since the problem demands Calories, Protein, Carbohydrate and Fat to be at least 500, 6, 10 and 8 respectively. Column H gives the required nutrient content.

	A	B	C	D	E	F	G	H	I	J
1	Decision Variables									
2		Food Item 1	Food Item 2	Food Item 3	Food Item 4					
3	Eaten	3	0	1	0					
4										
5	Objective Function									
6		Food 1	Food 2	Food 3	Food 4					
7	Eaten	3	0	1	0					
8	Cost per unit	0.5	0.2	0.3	0.8					
9										
10	Total Cost	1.8	=Sumproduct (B7:E7, B8:E8)				=Sumproduct(\$B\$7:\$E\$7,B13:E13)			
11										
12		Food Item 1	Food Item 2	Food Item 3	Food Item 4	Total		Required		
13	Calories	400	200	150	500	1350	≥	500		
14	Protein	3	2	0	0	9	≥	6		
15	Carbohydrates	2	2	4	4	10	≥	10		
16	Fat	2	4	1	5	7	≥	8		

- Step 4: Now, we will enter the Linear program into the solver. Now, once you have installed OpenSolver. When you click on the Data tab, on the right you will see Model. Click on the model, then enter the values one by one. First, we will enter the objective function, \$B10 i.e. in the objective cell. Select minimize because we want to minimize the diet cost.

OpenSolver - Model

What is AutoModel? AutoModel

AutoModel is a feature of OpenSolver that tries to automatically determine the problem you are trying to optimise by observing the structure of the spreadsheet. It will turn its best guess into a Solver model, which you can then edit in this window.

Objective Cell: ☐ maximise ☒ minimise ☐ target value:

Variable Cells:

Constraints:

\$F\$13 >= \$H\$13
 \$F\$14 >= \$H\$14
 \$F\$15 >= \$H\$15
 \$F\$16 >= \$H\$16

☒ Make unconstrained variable cells non-negative
☒ Show named ranges in constraint list

Sensitivity Analysis ☐ List sensitivity analysis on the same sheet with top left cell:
☐ Output sensitivity analysis: ☒ updating any previous output sheet ☐ on a new sheet

Solver Engine: Current Solver Engine: CBC

☒ Show model after saving

- Step 7: Now, you have to enter one important constraint. The non-negativity restriction. All the decision variables will be greater than 0.

- Step 9: Once the model is saved click on the Data tab then click solve. The optimal solution and values are displayed in the corresponding cells. The optimal minimum cost is US\$0.90. Sara should consume 3 units of Food Item 2 and 1 unit of Food Item 3 for the required nutrient content at the minimum cost. This solves our linear program.

	A	B	C	D	E	F	G	H	I
1	Decision Variables								
2		Food Item 1	Food Item 2	Food Item 3	Food Item 4				
3	Eaten	3	0	1	0				
4									
5	Objective Function								
6		Food 1	Food 2	Food 3	Food 4				
7	Eaten	0	3	1	0				
8	Per unit Cost	0.5	0.2	0.3	0.8				
9									
10	Total Cost	min 0.9							
11									
12		Food Item 1	Food Item 2	Food Item 3	Food Item 4	Total		Required	
13	Calories	400	200	150	500	750	≥	500	
14	Protien	3	2	0	0	6	≥	6	
15	Carbohydrates	2	2	4	4	10	≥	10	
16	Fat	2	4	1	5	13	≥	8	
17									

Case Study: Simplex Method

Simplex Method is one of the most powerful & popular methods for linear programming. The simplex method is an iterative procedure for getting the most feasible solution. In this method, we keep transforming the value of basic variables to get maximum value for the objective function.

A linear programming function is in its standard form if it seeks to maximize the objective function.

$$Z = c_1 x_1 + c_2 x_2 + \dots + c_n x_n \quad \text{subject to constraints,}$$

$$a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n \leq b_1$$

$$a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n \leq b_2$$

$$a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mn} x_n \leq b_m$$

where, $x_i \geq 0$ and $b_i \geq 0$. After adding slack variables, the corresponding system of constraint equation is,

$$a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n + s_1 = b_1$$

$$a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n + s_2 = b_2$$

$$a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mn} x_n + s_m = b_m \quad \text{where, } s_i \geq 0$$

The variables, $s_1, s_2, \dots, s_m$ are called slack variables. They are non-negative numbers that are added to remove the inequalities from an equation.

The above explanation gives the theoretical explanation of the simplex method. Now, I am gonna explain how to use the simplex method in real life using Excel.

Example: The advertising alternatives for a company include television, newspaper and radio advertisements. The cost for each medium with its audience coverage is given below.

Television	Newspaper	Radio	
Cost per advertisement (\$)	2000	600	300
Audience per advertisement	100,000	40,000	18,000

The local newspaper limits the number of advertisements from a single company to ten. Moreover, in order to balance the advertising among the three types of media, no more than half of the total number of advertisements should occur on the radio. And at least 10% should occur on television. The weekly advertising budget is \$18,200. How many advertisements should be run in each of the three types of media to maximize the total audience?

Solution: First I am going to formulate my problem for a clear understanding.

Step 1: Identify Decision Variables

Let X_1 , X_2 , X_3 represent the total number of ads for television, newspaper, and radio respectively.

Step 2: Objective Function

The objective of the company is to maximize the audience. The objective function is given by:

$$Z = 100,000X_1 + 40,000X_2 + 18,000X_3$$

Step 3: Write down the constraints

Now, I will mention each constraint one by one.

It is clearly given that we have a budget constraint. The total budget which can be allocated is \$18,200. And the individual costs per television, newspaper and radio advertisement is \$2000, \$600 and \$300 respectively. This can be represented by the equation,

$$2000X_1 + 600X_2 + 300X_3 \leq 18,200$$

For a newspaper advertisement, there is an upper cap on the number of advertisements to 10. My first constraints are, $X_2 \leq 10$

The next constraint is the number of advertisements on television. The company wants at least 10% of the total advertisements to be on television. So, it can be represented as:

$$X_1 \geq 0.10(X_1 + X_2 + X_3)$$

The last constraint is the number of advertisements on the radio cannot be more than half of the total number of advertisements. It can be represented as

$$X_3 \leq 0.5(X_1 + X_2 + X_3)$$

Now, I have formulated my linear programming problem. We are using the simplex method to solve this. I will take you through the simplex method one by one.

To reiterate all the constraints are as follows. I have simplified the last two equations to bring them in

standard form.

$$\begin{aligned} 2000X_1 + 600X_2 + 300X_3 &\leq 18,200 & X_2 &\leq 10 \\ -9X_1 + X_2 + X_3 &\leq 0 & -X_1 - X_2 + X_3 &\leq 0 \end{aligned}$$

We have a total of 4 equations. To balance out each equation, I am introducing 4 slack variables, S_1 , S_2 , S_3 and S_4 .

So equations are as follows:

$$\begin{aligned} 2000X_1 + 600X_2 + 300X_3 + S_1 &= 18,200 & X_2 + S_2 &= 10 \\ -9X_1 + X_2 + X_3 + S_3 &= 0 & -X_1 - X_2 + X_3 + S_4 &= 0 \end{aligned}$$

Solving these equations, the values for $X_1=4$, $X_2=10$ and $X_3=14$.

On solving the objective function the maximum weekly audience as 1,052,000.

Applications of Linear Programming

Linear programming and Optimization are used in various industries. The manufacturing and service industry uses linear programming on a regular basis. In this section, we are going to look at the various applications of Linear programming.

1. **Supply Chain Operations:** Manufacturing industries use linear programming for analyzing their supply chain operations. Their motive is to maximize efficiency with minimum operation cost. As per the recommendations from the linear programming model, the manufacturer can reconfigure their storage layout, adjust their workforce and reduce the bottlenecks. Here is a small Warehouse case study of Cequent a US-based company, watch this video for a more clear understanding.
2. **Shelf Space Optimization:** Linear programming is also used in organized retail for shelf space optimization. Since the number of products in the market has increased in leaps and bounds, it is important to understand what does the customer want. Optimization is aggressively used in stores like Walmart, Hypercity, Reliance, Big Bazaar, etc. The products in the store are placed strategically keeping in mind the customer's shopping pattern. The objective is to make it easy for a customer to locate & select the right products. This is subject to constraints like limited shelf space, a variety of products, etc.
3. **Optimizing Delivery Routes:** Optimization is also used for optimizing Delivery Routes. This is an extension of the popular traveling salesman problem. The service industry uses optimization for finding the best route for multiple salesmen traveling to multiple cities.

With the help of clustering and greedy algorithm, the delivery routes are decided by companies like FedEx, Amazon, etc. The objective is to minimize the operation cost and time.

4. **Optimizing Machine Learning Models:** Optimizations are also used in Machine Learning. Supervised Learning works on the fundamental of linear programming. A system is trained to fit a mathematical model of a function from the labeled input data that can predict values from unknown test data.